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- A weighted undirected graph with 8 nodes (A, B, C, D, E, F, G, H) and 11 edges. The weights are: A-B: 6, A-D: 6, B-C: 5, B-E: 10, C-D: 6, C-F: 7, D-H: 4, E-F: 6, F-G: 3, F-H: 6, H-G: 3. The total weight is 69.

[illegible]

Final Path	
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(b) Describe the purpose of both the A* algorithm and Dijkstra's algorithm.

[2]

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2. The table shows part of the instruction set for a processor which has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Op code	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC.
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC.
LDI	<address>	Indirect addressing. The address to be used is at the given address. Load the contents of this second address to ACC.
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the Index Register. Copy the contents of this calculated address to ACC.
LDR	#n	Immediate addressing. Load the number n to IX.
STO	<address>	Store the contents of ACC at the given address.
ADD	<address>	Add the contents of the given address to the ACC.
INC	<register>	Add 1 to the contents of the register (ACC or IX).
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX).
JMP	<address>	Jump to the given address.
CMP	<address>	Compare the contents of ACC with the contents of <address>.
CMP	#n	Compare the contents of ACC with number n.
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True.
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False.
AND	#n	Bitwise AND operation of the contents of ACC with the operand.
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>.
XOR	#n	Bitwise XOR operation of the contents of ACC with the operand.
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>.
OR	#n	Bitwise OR operation of the contents of ACC with the operand.
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>. <address> can be an absolute address or a symbolic address.
LSL	#n	Bits in ACC are shifted n places to the left. Zeros are introduced on the right hand end.
LSR	#n	Bits in ACC are shifted n places to the right. Zeros are introduced on the left hand end.
IN		Key in a character and store its ASCII value in ACC.
OUT		Output to the screen the character whose ASCII value is stored in ACC.
END		Return control to the operating system.

CS 9618 - Mock 2025

Write the assembly language program code that represents the following pseudocode algorithm.

```

COUNT ← 0
OUTPUT B
WHILE COUNT <> 5
    OUTPUT B
    COUNT ← COUNT + 1
ENDWHILE
OUTPUT A
END

```

Label	Op code	Operand	Comment
			// load 0
			// store in COUNT
			// load B
			// output B
LOOP:			// load COUNT
			// is COUNT = 5 ?
			// jump to ENDLOOP if TRUE
			// load B
			// output B
			// load COUNT
			// increment ACC
			// store in COUNT
			// jump to LOOP
ENDLOOP:			// load A
			// output A
			// end program
A:	1		
B:	0		
COUNT:			

[4]

3.

- (a) Given the following code snippet, analyze the order of time complexity and explain your reasoning. Assign the appropriate Big O notation to represent the time complexity.

Code Snippet:

```
FOR i ← 1 TO n
  FOR j ← 1 TO n
    OUTPUT "Hello, ZAK!"
  NEXT
NEXT
```

Big O notation:

Brief explanation:

..... [3]

- (b) Compare the performance of the algorithms for a binary search and a linear search using Big O notation for order of time complexity.

..... [3]

4. An international conference categorizes its submissions using codes generated according to specific rules. These rules are defined using the Backus-Naur Form (BNF), outlined as follows:

```
 <Category> ::= A|B|C|D|E
 <Year> ::= '20' <Digit><Digit>
 <Digit> ::= 0|1|2|3|4|5|6|7|8|9
 <SubmissionCode> ::= <Category><Year><Digit><Digit><Digit>
```

- (a) Create a definition for SubmissionCode as a record structure UDT, including components for Category, Year, and a sequence number.:

```
Record SubmissionCode
  DECLARE Category: _____
  DECLARE Year: _____
  DECLARE SequenceNumber: INTEGER
End Record
```

[2]

(b) State whether each of the following is a valid code or an invalid code. In the case of an invalid code, explain how the BNF rules given in question 4. are broken.

(i) A2023012 [2]

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(ii) C201875 [2]

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(c) The conference introduces a new code system to differentiate submissions.
Alter and/or extend the BNF definition given in 4 above to:

Define <ExtendedSubmissionCode> that includes an **optional** abstract prefix
"ABS-" before the <Category>. [4]

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5. The following statements pertain to a university's course enrollment system, described using a predicate logic based programming language where facts and rules state what to do but not how to do it.

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01 completed(zak,math101,a) .
02 completed(mike,math101,c) .
03 completed(sara,math101,b) .
04 completed(alex,math101,d) .
05 hasLicence(zak) .
06 hasLicence(sara) .
07 passedTheoryTest(zak,compsci201) .
08 passedTheoryTest(sara,compsci201) .
09 interestedInCourse(zak,compsci201) .
10 interestedInCourse(mike,compsci201) .
11 interestedInCourse(sara,compsci201) .
12 interestedInCourse(alex,compsci201) .
13 canEnroll(X,C)
    IF interestedInCourse(X,C) AND hasLicence(X) AND
    passedTheoryTest(X,C) .
```

Given the facts and rules in the code and references to clauses above:

- a. Give the name for type of programming language used . [1]

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- b. Give one example of a fact. [1]

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- c. Give one example of a rule. [1]

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- d. Give one example of a goal to query the given knowledge base. [1]

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- e. Give one example of resulting instantiation against the goal in d. above. [1]

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f. Create a rule that identifies students who are not eligible to enroll in a specific course because they:

-  either have not passed the required theory test for the course,
-  or do not have a license.

ineligibleForEnrollment (X,C)

IF

 [3]

6. In any calculations, you must show how you obtain your answers.
 A binary number can also be written in a floating point representation as:

0	1	0	1	0	0	1	0
mantissa					exponent		

(a) (i) Write down the denary value represented by the **exponent** in this example. [1]

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(ii) Calculate the denary value of the binary number in this floating point representation. [2]

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(b) Write, in binary, the largest number that can be represented in this floating point representation. [2]

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(c) Another binary number is written in the same format as:

0	0	1	1	1	0	1	1
mantissa					exponent		

Write this FP number in the normalised form.

[2]

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(d) State why binary numbers are normalised.

[1]

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7.

a. Explain why the number of machine cycles used by RISC and CISC to complete a single instruction is different.

[2]

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b. An array processor is used in SIMD systems.

(i) Explain the term **SIMD** processor.

[2]

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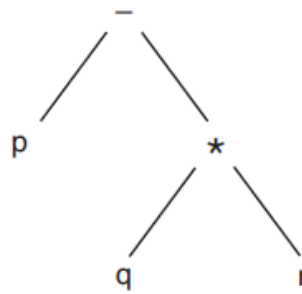
(ii) Give one example of the type of computer used, for which a **SIMD** processor is most suitable.

[1]

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- c. Following binary tree describes operands p , q and r with operators $-$ and $*$.



- (i) Write the reverse polish notation based on the binary tree above. [2]

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- (ii) State the type of traversal that should be used to obtain the reverse polish notation from a binary tree. [1]

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- (iii) An expression in reverse polish notation is stated as: $t \ u \ * \ v \ +$
 Showing how you obtained your answer, evaluate this expression, where:

$$T = 2, u = 3 \text{ and } v = 10 \quad [2]$$

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- (iv) State data structure, other than binary tree, you used to evaluate RPN in c(iii). [1]

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- (v) State a symbol, often used for infix mathematical notations, that is not required when working with RPN. [1]

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8. Complete truth table below for AND gate and draw the symbol for the same. [2]

(a)

Input A	Input B	Output

AND gate symbol

- (b) Table below list six Boolean laws. **Three** of them are correct. Shade the circles next to the three equations that are correct. [3]

Equation	Correct? (Shade three)
$A \cdot \bar{A} = 1$	☐
$A + B = \bar{\bar{A}} \cdot \bar{\bar{B}}$	☐
$A + 1 = 1$	☐
$A \cdot (A + B) = A$	☐
$A + (A \cdot B) = B$	☐
$A \cdot 1 = 1$	☐

- (c) Using the laws describing Boolean algebra, Simplify the following Boolean expression. [3]

$$(X + Y) \cdot (X + \bar{Y})$$

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9. The TCP/IP protocol suite can be viewed as a stack of layers.
a. (i) Complete the stack by inserting the names of the missing layers below.

Transport

[3]

(ii) State how each layer of the stack is implemented.

[1]

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b. A computer is currently running two processes:

 Process 1 is downloading and opening a web page in browser software.

 Process 2 is downloading an email using email client software.

Describe two tasks that the Transport layer performs to ensure that the incoming data is downloaded correctly.

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c. Name a possible protocol that will be used by the Process 1.

[1]

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d. Name a protocol that will be used by Process 2.

[1]

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